Scholarship Management System

(Case Study 61 | C++ Final Project)

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Problem Statement

🧩 Problem

Scholarship processing involves manual eligibility checks and fund tracking, causing delays and errors.

🎯 Objectives

Streamline scholarship allocation and reduce processing errors using an automated system.

⚙️ Functional Requirements

Student eligibility verification, scholarship allocation, and real-time status tracking.

📦 Deliverables

Working C++ scholarship system with full reports and documentation.

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Case Study 61.Scholarship Management System

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Scholarship processing involves eligibility verification and fund tracking. This system automates scholarship management.

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- To streamline scholarship allocation
- To reduce processing errors

⚙️ Functional Requirements

- Student eligibility verification
- Scholarship allocation
- Status tracking

📦 Deliverables

- Scholarship system
- Reports and documentation

Our Solution

Scholarship Class

- name, minGPA, maxIncome, fundAmount
- checkEligibility(gpa, income)
- Returns status string

Student Class

- studentID, name, gpa, familyIncome
- scholarshipStatus, allocatedFunds
- displayStatus() method

ScholarshipSystem

- addStudent() with duplicate ID check
- processScholarships() – auto-match
- displayReport() + trackStudentByID()

💡 **Built-in Scholarships:**

Merit Base — GPA ≥ 8.5, Income ≤ ₹8,00,000 → Fund: ₹50,000

Need Base — GPA ≥ 7.0, Income ≤ ₹4,00,000 → Fund: ₹30,000

Working / Method

STEP 1

Add Student

User enters ID, name, GPA & income.
System validates input & checks for duplicate ID.

STEP 2

Process Scholarships

System compares each student's GPA & income
against all defined scholarship criteria.

STEP 3

Eligibility Check

checkEligibility() returns ELIGIBLE, 'GPA too low', 'Income exceeds limit', or both.

STEP 4

Allocation & Report

Approved students get funds assigned.
Full report and individual tracking available.

🛡️ Input Validation: getValidID() ensures positive integers | getValidGPA() enforces 0.0–10.0 range | getValidIncome() requires non-negative values — all use exception-safe parsing with try/catch.

Conclusion

Objective Achieved

- Fully automated scholarship
- eligibility check
- fund allocation
- eliminating manual errors.

OOP Principles Applied

Three well-structured classes —
- Scholarship,
- Student,
- ScholarshipSystem — with encapsulation.

Robust Validation

Input guards prevent duplicate IDs,
invalid GPA/income values with helpful error messages.

Future Scope

- File storage
- dynamic scholarship addition
- GUI interface
- advanced ranking algorithms.

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Thank You

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